System Design

Introduction

The system design phase marks a critical juncture in the development of the CodeQuest system. In this section, we delve into the architectural framework, components, and interactions that shape the foundation of our innovative solution.The design goal for our system is to provide a **Secure Code Execution Environment** and **Intelligent Feedback**.

- **Traceability**: The design realizes our goals for **Quest Evaluation** and **Secure Infrastructure**.
- **Constraints**: The architecture is dictated by the **5-second execution limit** and the need for **isolated Execution sandbox** to prevent security breaches, similar to how the reference project uses isolation for its ML models.

Current Software Architecture

Note: Since this is a new build, you can state: "As CodeQuest is a greenfield project, there is no legacy architecture being replaced."

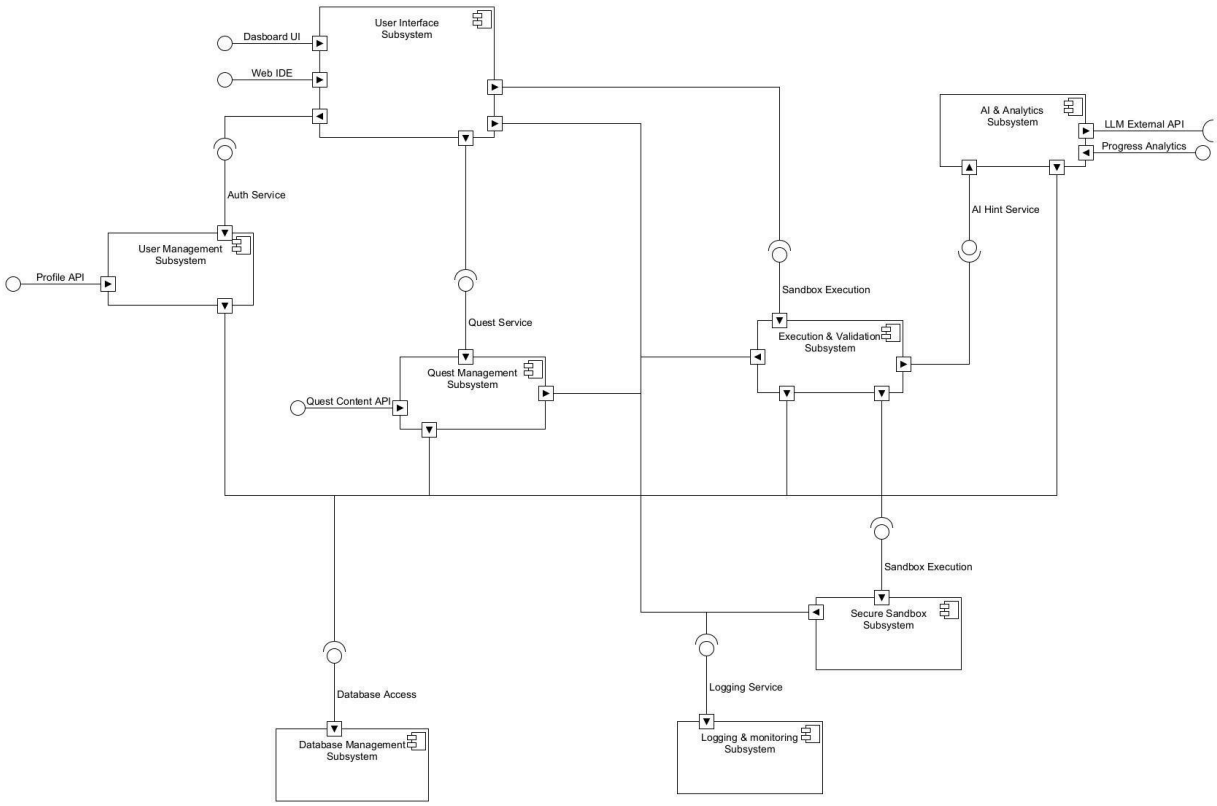
Proposed Software Architecture

Overview

Subsystem Decomposition

Subsystem decomposition is a process in a system design where the overall system is broken down into smaller, more manageable subsystems or modules. Each subsystem are designed to handle a specific set of functionalities, and together, they contribute to the overall functioning of the entire system.
This system is broken down into five main subsystems:

1. **User Interface (UI) Subsystem**
2. **User Management Subsystem**
3. **Quest Management Subsystem**
4. **Execution & Validation Subsystem**
5. **AI & Analytics Service Subsystem**
6. **Database Management Subsystem**
7. **Logging & Monitoring Subsystem**
8. **Secure Sandbox Subsystem**



User Interface (UI) Subsystem

manages the Web IDE, the Progress Dashboard, and the Admin portal.

User Management Subsystem

Handles authentication and profile analytics.

Quest Management Subsystem

Admin tool for managing challenges and test cases.

Execution & Validation Subsystem

Interfaces with the Sandbox to run code and validates results.

AI & Analytics Service Subsystem

Connects to an external LLM for personalized hint generation & Process analytics as well progress

Database Management Subsystem

Stores persistent records of quests, users, and progress.

Logging & Monitoring Subsystem

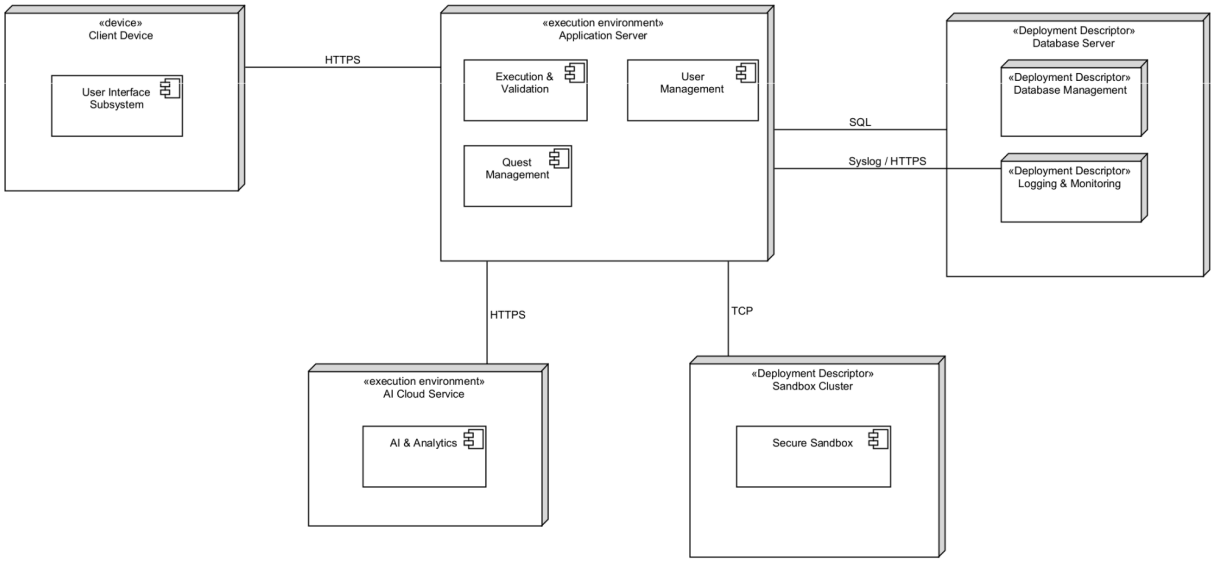
Handles logging unauthorized access attempts, and tracking system performance.

Secure Sandbox Subsystem

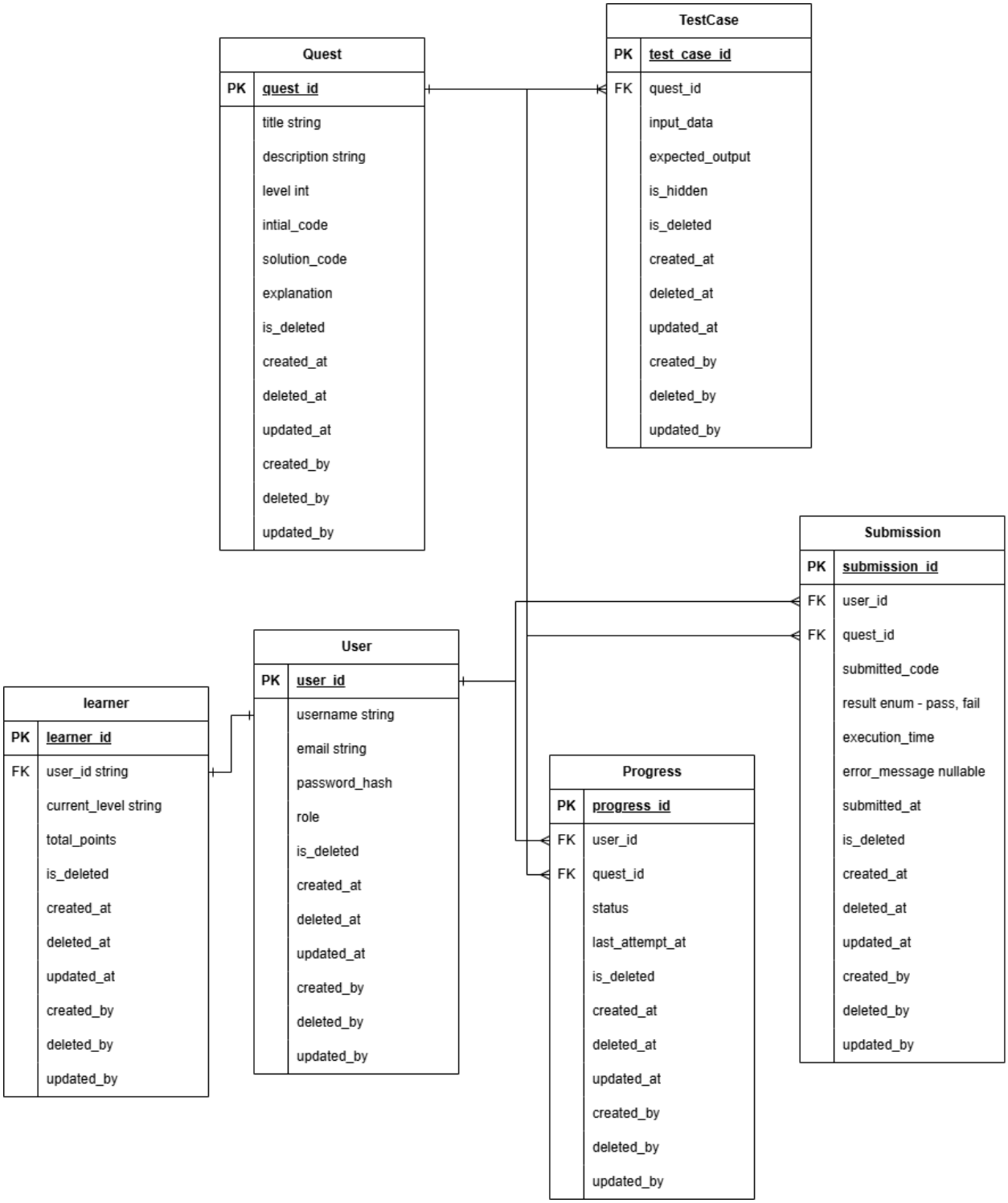
Handles security for the containers, detects and terminates containers after viewing suspicious logs

Hardware/Software Mapping

To prevent I/O contention on the Database Server node, the system is configured to use independent storage volumes for the relational database and the system logs.



Persistent Data Management



Access Control and Security

Access control determines which functionalities of the system each user is granted permission to access. Different users have varying levels of access to different subsystems within the system. To ensure security and protect against unauthorized access, malicious data modifications, and accidental introduction of inconsistencies, the system implements access control policies and security measures.

Subsystem	Components	Operation	Users